

CS57300
PURDUE UNIVERSITY

DATA MINING

CONVEX FUNCTIONS

- ▶ Sum of convex functions is convex; max of convex functions is convex.

SOLVE CONVEX OPTIMIZATION PROBLEM

- ▶ Minimize a convex function without any constraints on the variables
 - ▶ If $f'(x)=0$ then x is a stationary point of f
 - ▶ If $f'(x)=0$ and $f''(x)$ is not negative then x is a local minimum of f (for convex function, this is also a global minimum)
 - ▶ If f is a strictly convex function, any stationary point of f is the unique global minimum of f
- ▶ What about minimizing a convex function with constraints?

USE LAGRANGE MULTIPLIERS TO SOLVE CONVEX OPTIMIZATION

- ▶ For a standard form of convex optimization problem (f_0 and f_i are convex, h_i is linear):

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad \text{for } i = 1, \dots, m. \\ & h_i(x) = 0, \quad \text{for } i = 1, \dots, k.\end{array}$$

- ▶ The Lagrangian function of it is

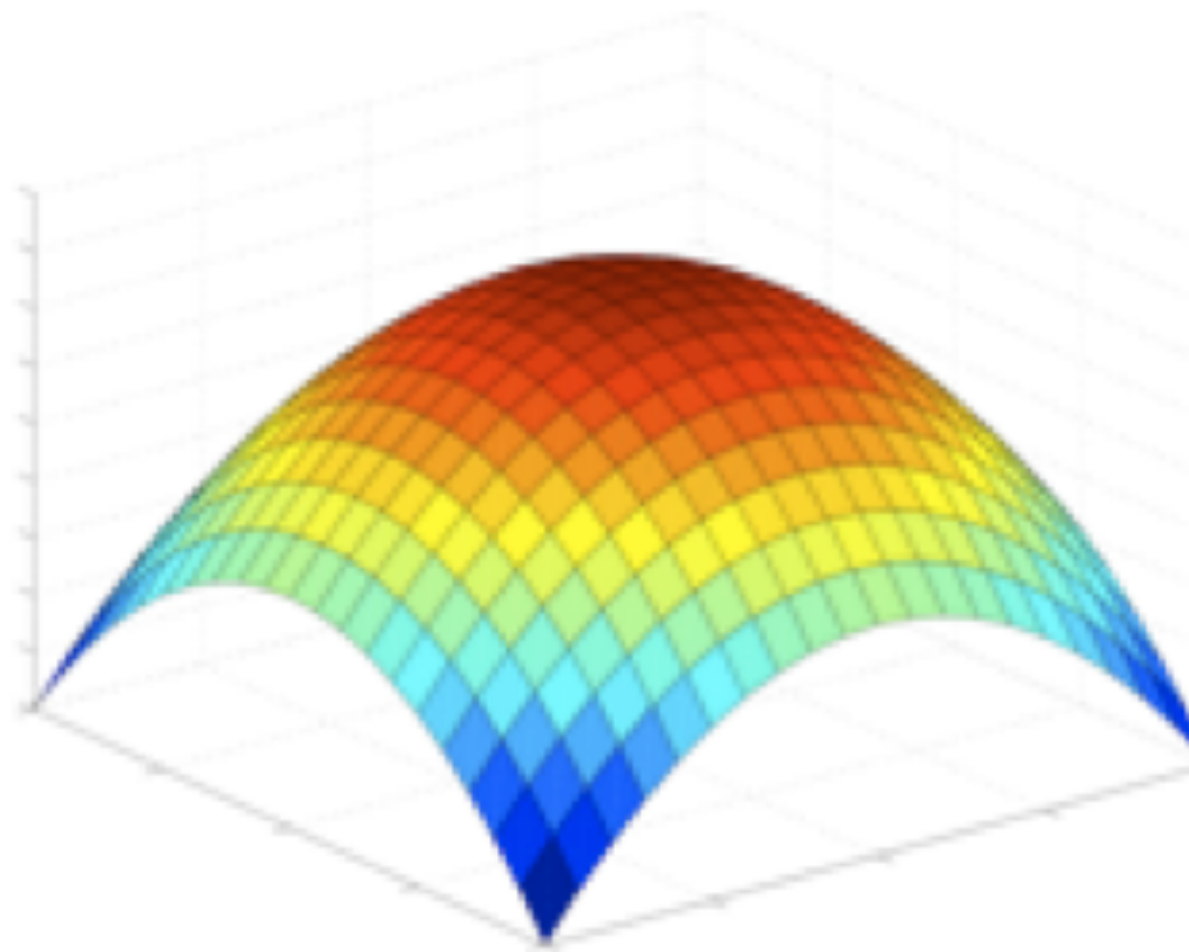
$$L(x, \lambda, v) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^k v_i h_i(x)$$

- ▶ $\lambda_i \geq 0$ is the **Lagrange multiplier** for the i -th inequality constraint, v_i is the **Lagrange multiplier** for the i -th equality constraint
- ▶ Solve the constrained optimization problem by finding the stationary point of the Lagrangian function

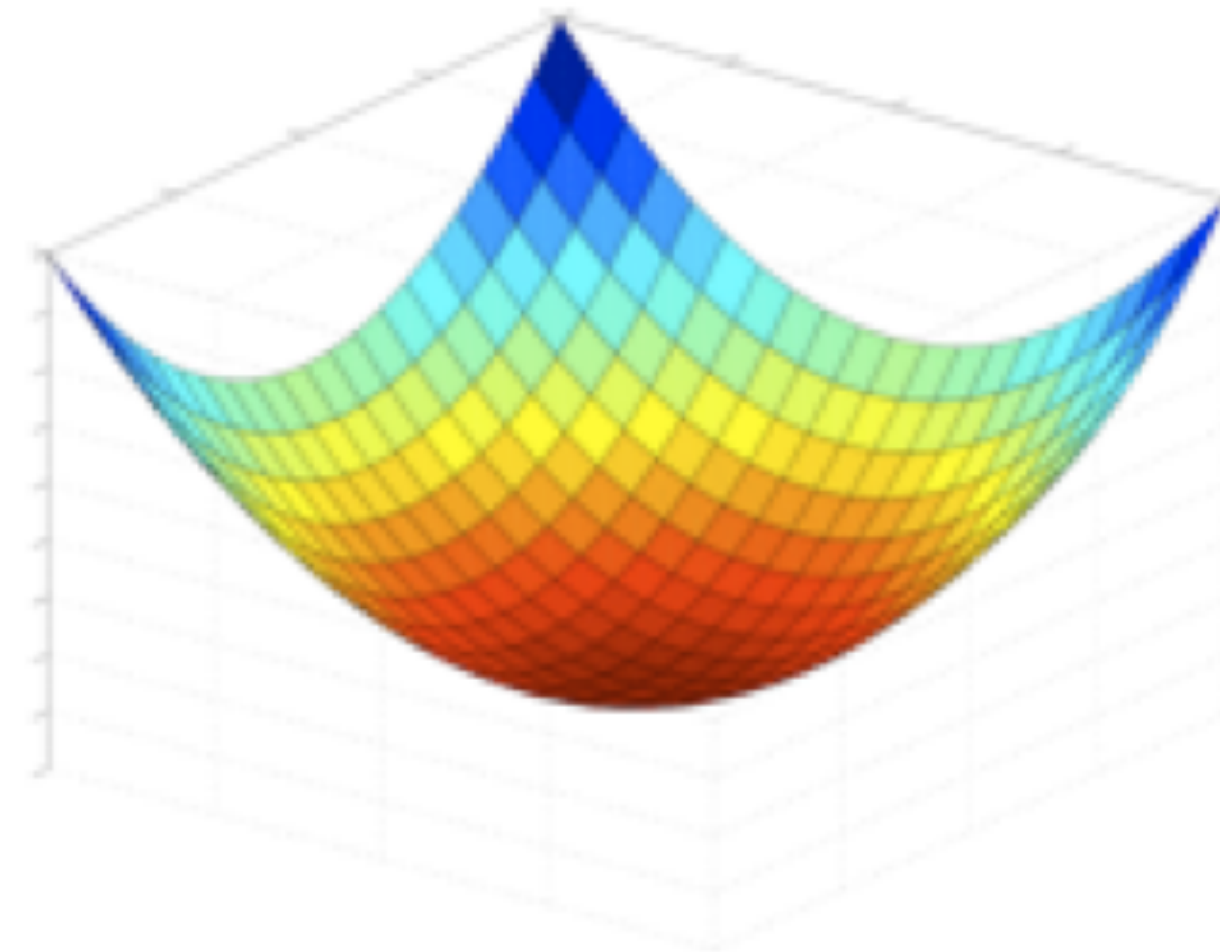
CONCAVE VS CONVEX

- ▶ Maximizing a concave function is equivalent to minimizing a convex function

concave



convex



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- ▶ Maximize the log likelihood function

- ▶ Likelihood:
$$L(\theta | D) = \prod_{i=1}^n \prod_{j=1}^m P(x_{ij} | c_i) P(c_i) = \left(\prod_{l=1}^L p_l^{N_l} \right) \left(\prod_{l=1}^L \prod_{j=1}^m \prod_{k=1}^{K(j)} (q_l^{jk})^{N_l^{jk}} \right)$$

- ▶ Log Likelihood:
$$\log L(\theta | D) = \sum_{l=1}^L N_l \log(p_l) + \sum_{l=1}^L \sum_{j=1}^m \sum_{k=1}^{K(j)} N_l^{jk} \log q_l^{jk}$$

- ▶ Subject to constraints:
$$\sum_{l=1}^L p_l = 1, \sum_{k=1}^{K(j)} q_l^{jk} = 1$$

- ▶ Lagrangian function:

$$L(p_l, q_l^{jk}, v_0, v_{lj}) = \sum_{l=1}^L N_l \log(p_l) + \sum_{l=1}^L \sum_{j=1}^m \sum_{k=1}^{K(j)} N_l^{jk} \log q_l^{jk} + v_0 \left(\sum_{l=1}^L p_l - 1 \right) + \sum_{l=1}^L \sum_{j=1}^m v_{lj} \left(\sum_{k=1}^{K(j)} q_l^{jk} - 1 \right)$$

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$$L(p_l, q_l^{jk}, v_0, v_{lj}) = \sum_{l=1}^L N_l \log(p_l) + \sum_{l=1}^L \sum_{j=1}^m \sum_{k=1}^{K(j)} N_l^{jk} \log q_l^{jk} + v_0 \left(\sum_{l=1}^L p_l - 1 \right) + \sum_{l=1}^L \sum_{j=1}^m v_{lj} \left(\sum_{k=1}^{K(j)} q_l^{jk} - 1 \right)$$



$$\frac{N_l}{p_l} + v_0 = 0, \frac{N_l^{jk}}{q_l^{jk}} + v_{lj} = 0$$



$$p_l = -\frac{N_l}{v_0}, q_l^{jk} = -\frac{N_l^{jk}}{v_{lj}} \quad + \quad \sum_{l=1}^L p_l = 1, \sum_{k=1}^{K(j)} q_l^{jk} = 1$$



$$p_l = \frac{N_l}{N}, q_l^{jk} = \frac{N_l^{jk}}{N_l}$$

GRADIENT DESCENT

- ▶ For some convex functions, we may be able to take the derivative, but it may be difficult to directly solve for parameter values
- ▶ Solution:
 - ▶ Start at some value of the parameters
 - ▶ Take derivative and use it to move the parameters in the direction of the negative gradient
 - ▶ Repeat until stopping criteria is met (e.g., gradient close to 0)

Gradient Descent Rule:

$$\underline{\mathbf{w}}_{\text{new}} = \underline{\mathbf{w}}_{\text{old}} - \eta \Delta(\underline{\mathbf{w}})$$

where

$\Delta(\underline{\mathbf{w}})$ is the gradient and
 η is the learning rate (small, positive)

Notes:

1. This moves us downhill in direction $\Delta(\underline{\mathbf{w}})$ (steepest downhill direction)
2. How far we go is determined by the value of η

ILLUSTRATION OF GRADIENT DESCENT

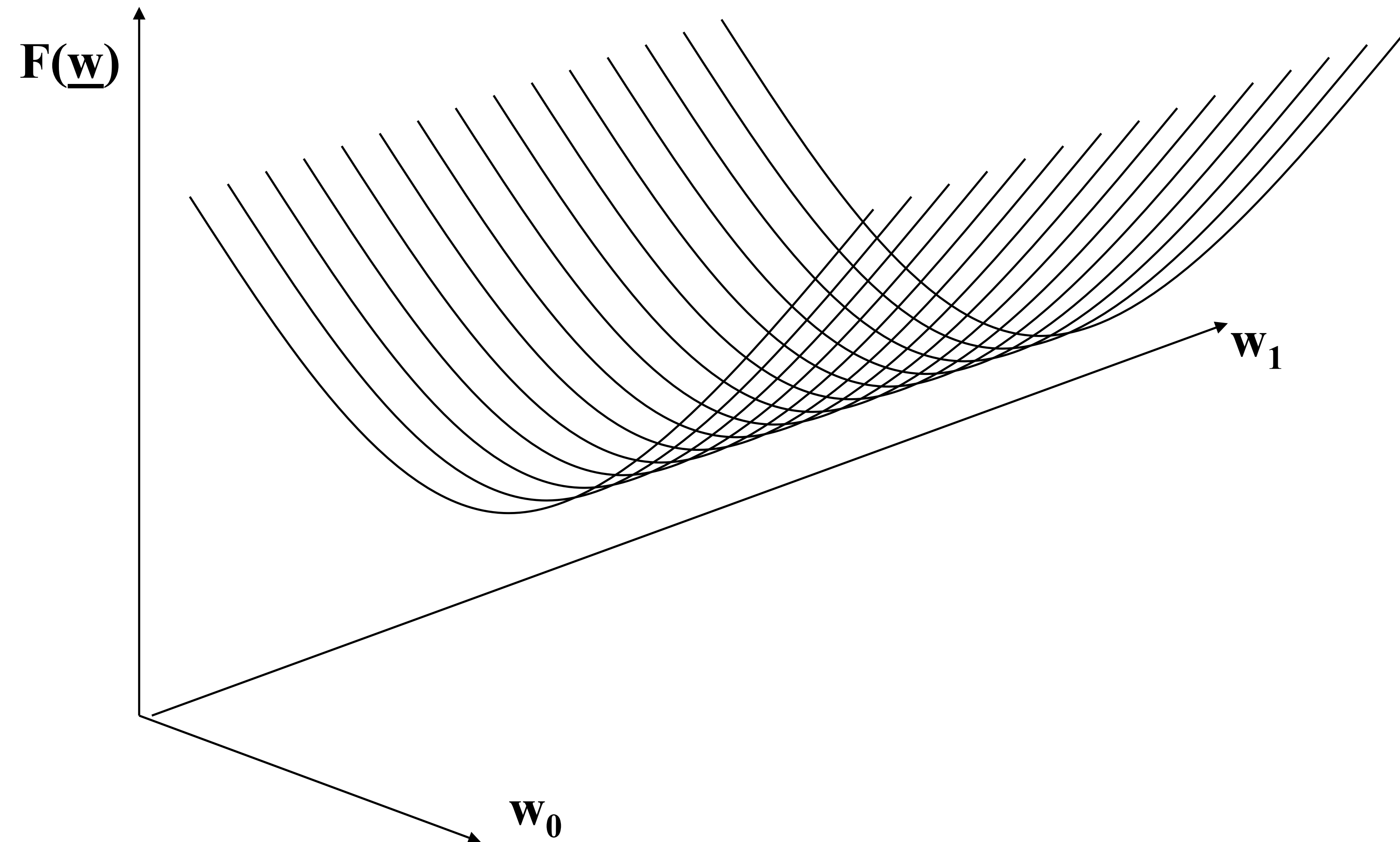


ILLUSTRATION OF GRADIENT DESCENT

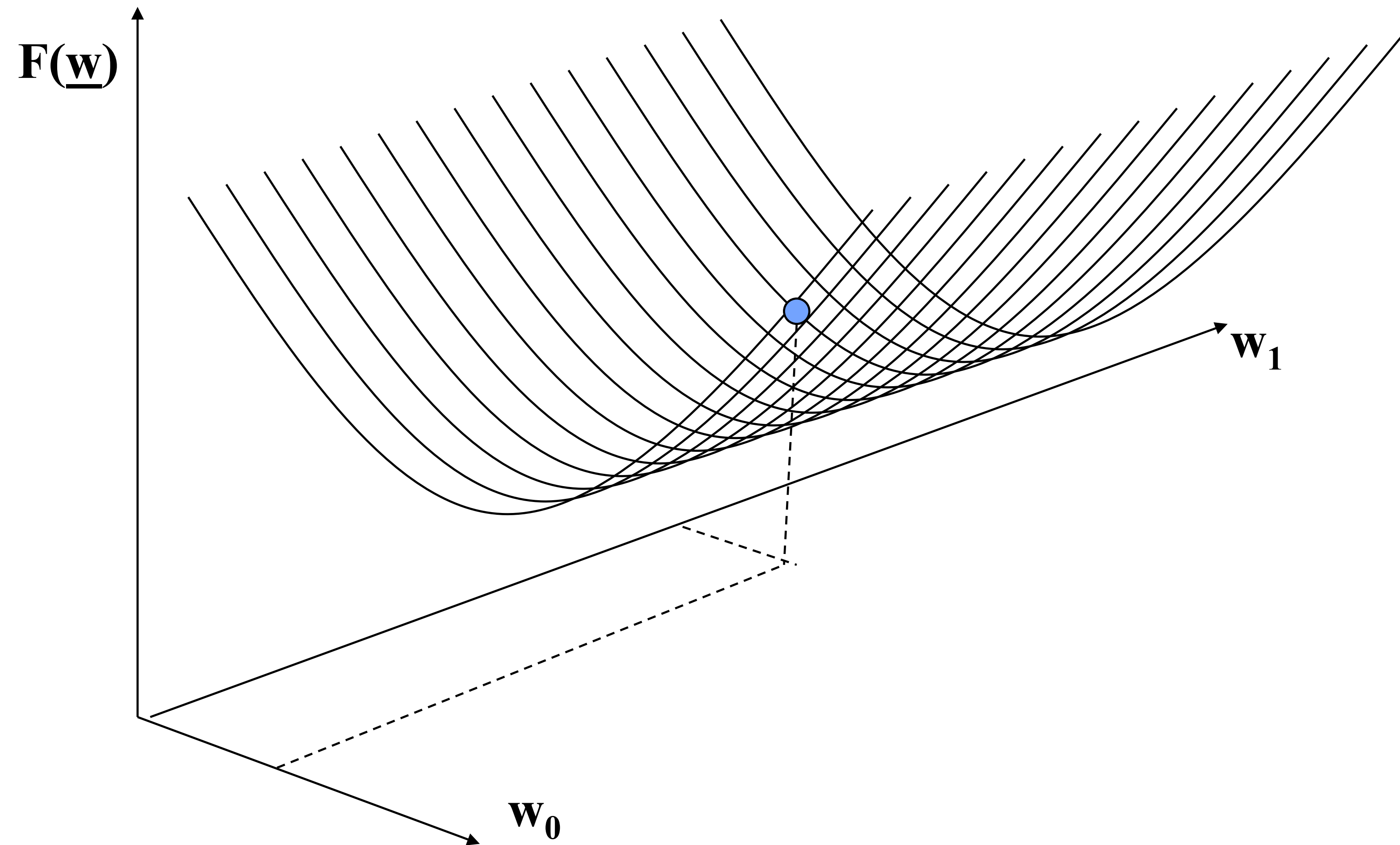
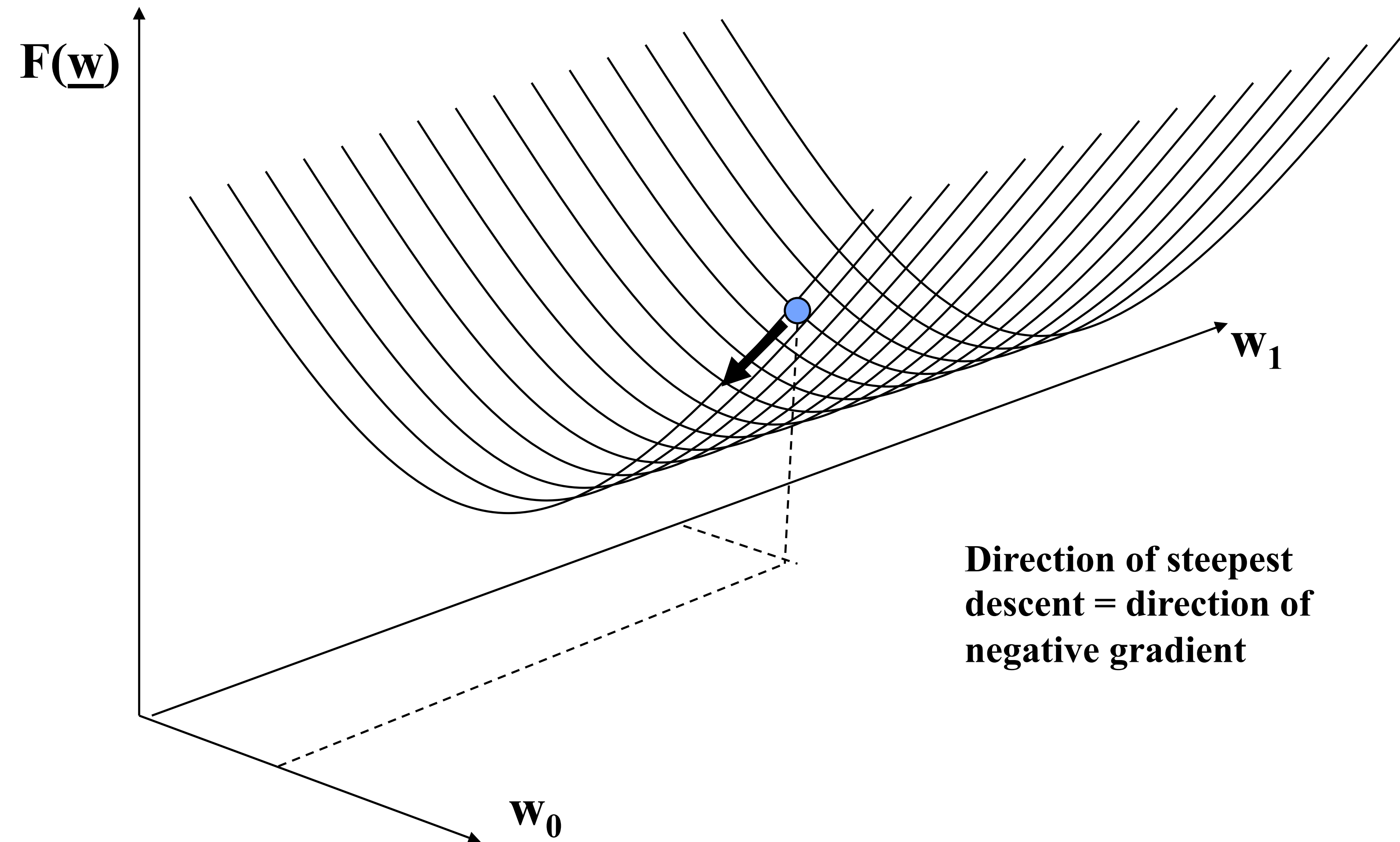
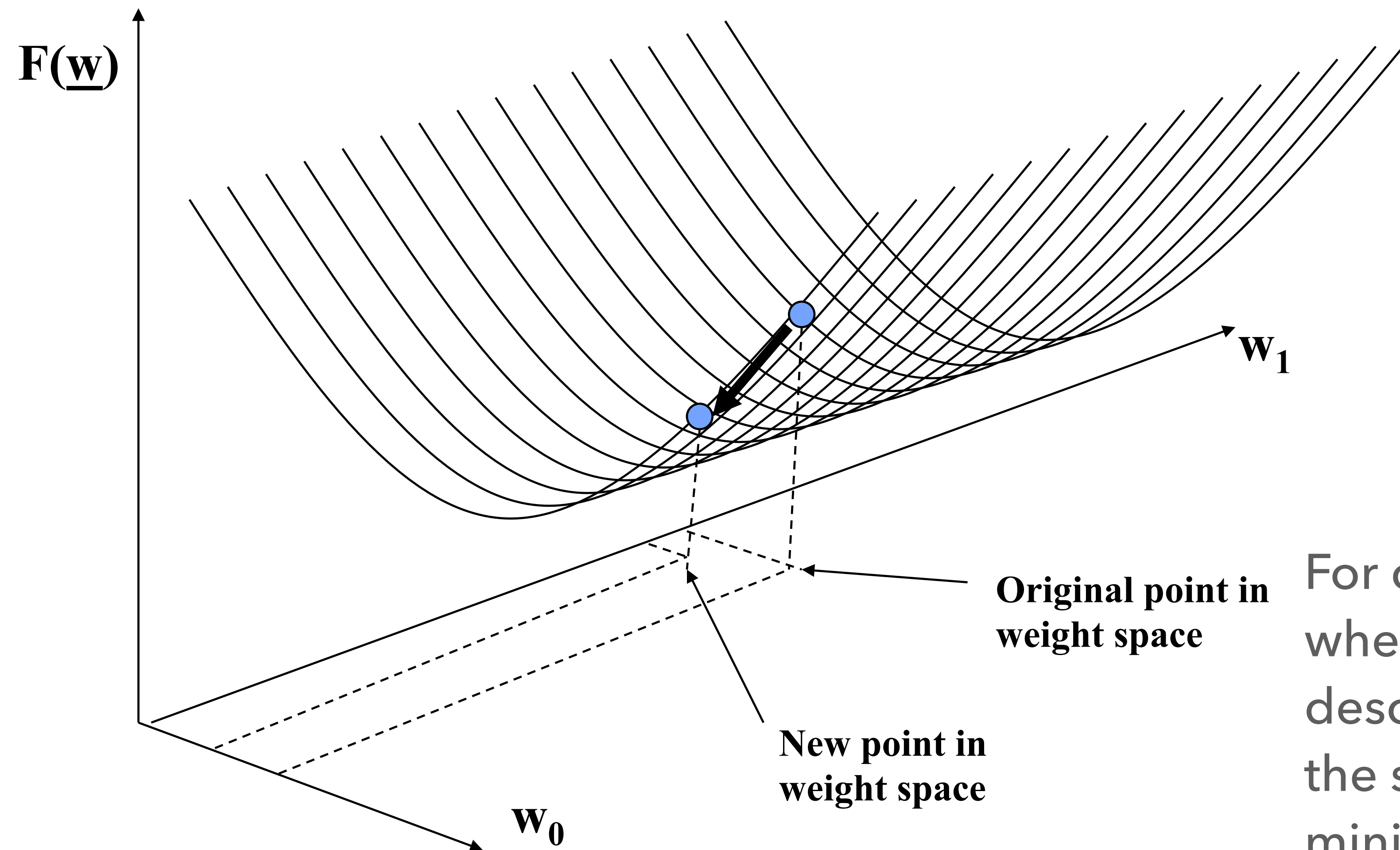


ILLUSTRATION OF GRADIENT DESCENT



**Direction of steepest
descent = direction of
negative gradient**

ILLUSTRATION OF GRADIENT DESCENT



For convex functions, when gradient descent converges, the solution is global minimum.

STOPPING CRITERIA FOR GRADIENT DESCENT

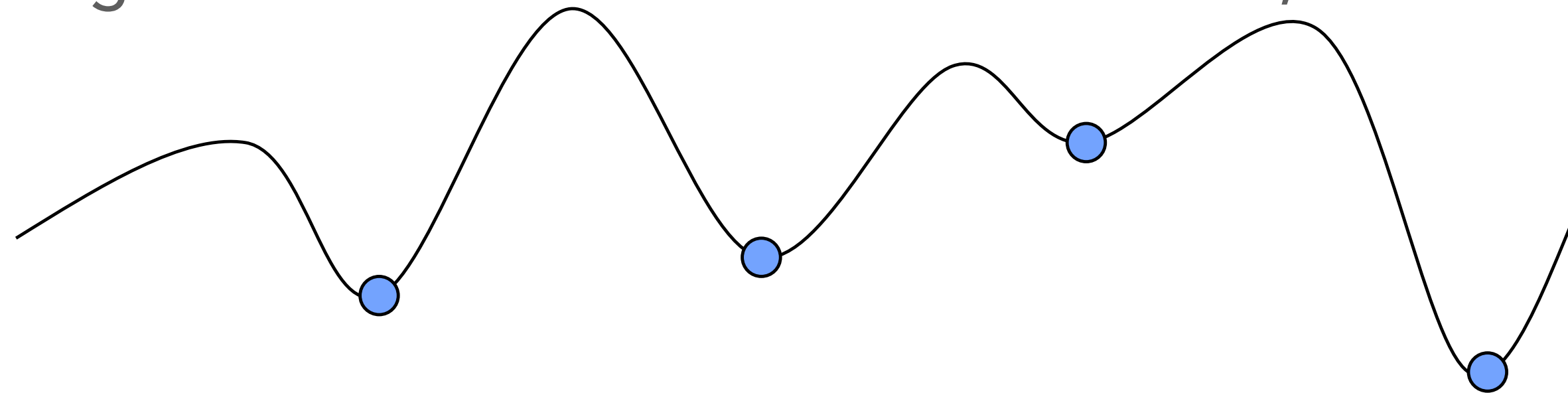
- ▶ Ideally, $f'(x)=0\dots$
- ▶ In practice...
 - ▶ $\|\nabla f(x)\| < \varepsilon$
 - ▶ $|f(x_{k+1}) - f(x_k)| < \varepsilon$
 - ▶ $\|x_{k+1} - x_k\| < \varepsilon$
 - ▶ Maximum number of iterations has been reached

GRADIENT ASCENT

- ▶ For concave functions that you want to *maximize*, take a step in direction of gradient (i.e., $w_{\text{new}} \leftarrow w_{\text{old}} + \eta \nabla(w)$)
- ▶ Otherwise same as gradient descent:
 - ▶ Start at some parameter values
 - ▶ Take derivative, move the parameters in the direction of gradient
 - ▶ Repeat until stopping criteria is met (e.g., gradient close to 0)

GRADIENT DESCENT FOR NON-CONVEX OPTIMIZATION

- ▶ Works on any objective function $F(\theta)$
 - ▶ as long as we can evaluate the gradient $\Delta(\theta)$
 - ▶ this can be very useful for minimizing complex functions F
- ▶ Can be used in hill-climbing search to find local minima in smooth, but non-convex functions



- ▶ If function has multiple local minima, gradient descent goes to the closest local minimum:
 - ▶ solution: random restarts from multiple places in model space